



TECHNICAL MAINTENANCE MANUAL

HVAC Unit Maintenance Manual - UH-HVAC-302

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Client	UrbanHive Facility Technologies Pvt. Ltd.
Site	GreenPark Meridian Residency, Whitefield, Bengaluru
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Review cycle	Annual or after any major incident

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1. Equipment Overview and Technical Specifications

Variable refrigerant flow (VRF) outdoor unit serving residential towers at GreenPark Meridian. R410A refrigerant circuit, inverter compressor, electronic expansion valves, and BACnet gateway to BMS.

Model code: UH-HVAC-302

Asset registry: Record serial number and installation tower in AMIS before first energization.

Electrical supply: 415V 3-phase supply, dedicated 40A MCB, surge protection Type 2, earth leakage 30mA

Environmental limits: Ambient 0-45°C operational; store spare parts below 35°C and 60% RH.

Standards: UrbanHive engineering bulletin UH-ENG-AMIS-2024; applicable Indian Standards as cited per asset class.

Documentation package: This manual, wiring diagram sheet WD-UH-HVAC-302, commissioning checklist CC-019, and spare parts catalogue SP-UH-HVAC-302.

Technicians must verify nameplate data matches AMIS asset record UH-HVAC-302 before applying procedures in this document. Discrepancies must be logged as a data quality ticket to UrbanHive Data Engineering within one business day.

2. Installation Guidelines

Mount on structural pedestal with minimum 300mm rear clearance for discharge airflow. Vacuum hold test 500 microns for 4 hours before charging. Charge per superheat/subcooling worksheet UH-HVAC-CHG-01.

Pre-commissioning checks:

- Verify incoming power quality: voltage balance within 2%, THD < 5% on dedicated feeder.
- Confirm earthing resistance ≤ 1 ohm at equipment location; bond to society main earth bar.
- Inspect shipping damage; photograph defects for vendor warranty within 48 hours.
- Complete mechanical alignment, torque logs, and insulation resistance tests per UrbanHive QC form UH-QC-INST-02.

Post-installation: Upload signed commissioning certificate, as-built photos, and warranty start date to AMIS. Tag physical asset with QR code linking to this manual revision v1.0.0.

3. Preventive Maintenance Schedule

UrbanHive mandates the following intervals for GreenPark Meridian. Adjust only with written approval from Head of Maintenance.

- Daily (summer): discharge pressure trend review via BMS.
- Weekly: condenser coil wash if $\Delta P > 15$ kPa; drain pan inspection on indoor units.
- Monthly: refrigerant leak sniff at flare joints; fan motor amp balance within 10%.
- Quarterly: compressor insulation test; oil level check on outdoor modules.
- Annual: full refrigerant analysis; replace drier core if moisture indicator pink.

Documentation: Each PM visit must generate a service log in AMIS with parts consumed, meter readings, and pass/fail checklist. Missed PM windows trigger automatic escalation to Facility Manager after 72 hours.

4. Troubleshooting Guide

Use isolation procedure: observe symptom ? confirm with meter/test ? apply remediation ? functional test ? document in AMIS.

If fault persists after two visits, escalate to UrbanHive vendor support with fault code and photos.

Fault 01 - Code H101

Symptom: Insufficient cooling - superheat high

Remediation: Verify filter drier restriction; check EEV opening signal; confirm indoor coil not iced.

Fault 02 - Code H208

Symptom: Compressor short-cycle

Remediation: Inspect high-pressure switch setting; verify condenser airflow; check refrigerant charge weight.

Fault 03 - Code H315

Symptom: E4 fault - discharge temp protection

Remediation: Clean condenser coil; verify fan operation; reduce setpoint ramp rate in BMS.

Fault 04 - Code H422

Symptom: Communication loss IDU-ODU

Remediation: Check field wiring polarity on S1/S2; terminate bus ends; inspect for 120? resistor.

Fault 05 - Code H530

Symptom: Refrigerant leak code L3

Remediation: Nitrogen pressure test at 35 bar; soap test flare nuts; recover and recharge per worksheet.

Fault 06 - Code H601

Symptom: Indoor unit water drip

Remediation: Clear drain line slope 1:100; flush with biocide; verify condensate pump float.

Fault 07 - Code H714

Symptom: Abnormal compressor current imbalance

Remediation: Megger windings; inspect terminal lug torque 12 Nm; check for liquid slugging.

Fault 08 - Code H802

Symptom: Outdoor fan inverter fault

Remediation: Reset after 3 minutes; inspect DC bus capacitors; verify phase supply symmetry.

Fault 09 - Code H905

Symptom: BMS BACnet offline

Remediation: Ping gateway IP; verify MSTP address unique; reload point map UH-BMS-HVAC-MAP.

Fault 10 - Code H1003

Symptom: Oil return deficiency alarm

Remediation: Run minimum 30% load for 45 minutes; check oil separator temperature.

Fault 11 - Code H1108

Symptom: Frost on suction line

Remediation: Replace filter drier; check expansion valve superheat target 6K.

Fault 12 - Code H1201

Symptom: Excessive noise > 68 dB(A)

Remediation: Tighten panel screws; inspect compressor anti-vibration mounts; verify fan blade balance.

5. Spare Parts List

Maintain minimum stock at GreenPark Meridian central store (Basement B1, Room ST-04).

Part No.	Description	Min stock
SP-H01	Filter drier 16 cu in	6
SP-H02	EEV coil assembly	4
SP-H03	Outdoor fan motor	2
SP-H04	Drain pan biocide tablets	20
SP-H05	BACnet gateway module	1
SP-H06	Refrigerant R410A 11.3kg	4

Reorder when stock falls below minimum. Use UrbanHive procurement portal PR-UH-OPS for approved vendors only.

6. Safety Warnings and Compliance

Only F-gas certified technicians may handle refrigerant. Recover before opening circuit. PPE: safety goggles, insulated gloves, refrigerant detector. Work at height harness for rooftop units.

Permit requirements: Hot work, confined space, work-at-height, and electrical isolation require digital permit in AMIS before start.

PPE minimum: Safety shoes IS 15298, helmet IS 2925, eye protection IS 5983, gloves appropriate to task.

Incident reporting: Any injury, near-miss, or property damage - complete IR-01 within 30 minutes; notify security desk ext. 9911.

7. Manufacturer Contact and UrbanHive Support

OEM technical support: +91-80-5555-0019 (08:00-20:00 IST)

Email: support-uh-hvac-302@urbanhive-vendor.example.com

UrbanHive Ops escalation: maintenance@urbanhiveops.example.com - Priority 1 response within 4 business hours

AMIS helpdesk: Log ticket category `HVAC Unit` with asset_id from tower registry

Emergency (life safety assets): 24x7 hotline +91-80-5555-0911

Revision history: v1.0.0 - Initial release for GreenPark Meridian AMIS integration and UrbanHive contractor certification programme.

End of document MAN-019